Dr. Gunjan D. Mehta, Ph.D.

Assistant Professor/Principal Investigator
Department of Biotechnology
Indian Institute of Technology Hyderabad, India
(M.) +91 70168 96886
Email: gunjanmehta@bt.iith.ac.in

EDUCATION

2009-2015	Ph.D. Cell and Molecular Biology	Indian Institute of Technology, Bombay, India
2007-2009	M.Sc. Microbiology	M.S. University of Baroda, Vadodara, India
2003-2006	B.Sc. Biochemistry/Biotechnology	St. Xavier's College, Ahmedabad, India

PROFESSIONAL EXPERIENCE

2020-Now Assistant Professor/Principal Investigator, Department of Biotechnology, Indian Institute of Technology Hyderabad, India
Project: Developing Single-Molecule Imaging approaches to understand the mechanism of cell division and gene regulation with implications for infertility, genetic disorders, and cancers

2015-2020 Post-Doctoral Fellow, National Cancer Institute (NCI), National Institutes of Health (NIH), Bethesda, MD, USA
Project: Understanding transcription initiation and chromatin remodelling at the single-molecule level and its implication in stress response and cancer research
Mentors: Dr. Tatiana Karpova/Dr. Gordon Hager

2009-2015 Ph.D., Department of Biosciences and Bioengineering, IIT Bombay, Mumbai, India
Project: Understanding the mechanism of meiotic chromosome segregation and kinetochore organization and their significance in aneuploidy, developmental disorders, and cancers
Mentor: Prof. Santanu Ghosh

2007-2009 M.Sc. Dissertation, Department of Microbiology and Biotechnology, M.S. University of Baroda, Vadodara, India
Project: Genetic engineering and statistical optimization of medium constituents for increased chitinase production for its applications in agriculture and medicine
Mentor: Dr. Pranav Vyas

2006-2007 Research Assistant, Fermentation R&D, Intas Biopharmaceuticals Ltd., Ahmedabad, India
Project: Process development and technology transfer for the production of human therapeutic proteins, cytokines and hormones
Mentor: Dr. Rashbehari Tunga

2004-2006 Research Trainee, Industrial Training Unit, St. Xavier's College, Ahmedabad, India
Project: Process development for the micropropagation of the ornamental plants.
Mentor: Dr. Vincent Briganza

PUBLICATIONS

After joining IIT Hyderabad:

1) Paliwal S, Dey P, Kumari A, Sadasivan K, Joshi S, Shinohara A, Nishant K.T., **Mehta G***. A dynamic pool of Rec8-cohesin is crucial for meiotic recombination and transcription regulation in the yeast *S. cerevisiae*. ***Journal of Biological Chemistry*** (Under review), IF: 4, Q1
2) Mann R, Soota D, Das A, Podh NK, **Mehta G**, Notani D*. Paused polymerase synergises with transcription factors and RNA to enhance the transcriptional potential of promoters. ***Nature Communications*** **2025 (Accepted)**, IF: 17.2, Q1
3) Podh NK, Das A, Kumari A, Garg K, Yadav R, Kashyap K, Islam S, Gupta A, **Mehta G***. Single-molecule tracking reveals the dynamic turnover of Ipl1 at the kinetochores in *S. cerevisiae*. ***Life Science Alliance*** **2025**; 8(7),e202503290, IF: 3.5, Q1
4) Reza MH, Aggarwal R, Verma J, Podh NK, Chowdhury R, **Mehta G**, Manjithaya R, Sanyal K*. Spindle pole body-associated Atg11, an autophagy-related protein, regulates microtubule dynamics essential for high-fidelity chromosome segregation. ***PLoS Biology*** **2025**; 23(4):e3003069. IF: 7.7, Q1
5) Kumari A, Podh NK, Sen S, Kashyap K, Islam S, Gupta A, Rajakumara E, Nambiar M, **Mehta GD***. Single-Molecule Tracking dataset for histone H3 (*hht1*) from live cells of *Schizosaccharomyces pombe*. ***Nature Scientific Data*** **2024**;11:1393, IF: 8.7, Q1
6) Paliwal S, Dey P, Tambat S, Shinohara A, **Mehta GD***. Role of ATP-dependent chromatin remodelers during meiosis. ***Trends in Genetics (Cell Press)*** **2024***;41(3):236-250,* IF: 16.3, Q1
7) Kinger S, Jagtap YA, Kumar P, Choudhary A, Prasad A, Prajapati VK, Kumar A, **Mehta G**, Mishra A. Proteostasis in neurodegerative diseases. ***Advances in Clinical Chemistry*** **2024**;121:270-333. IF: 6.3, Q2
8) Podh NK#, Das A#, Dey P, Paliwal S, **Mehta G***. Single-Molecule Tracking Dataset of Histone H3 (Hht1) in *Saccharomyces cerevisiae*. ***Data in Brief*** **2023**;47:108925, IF: 1.4, Q3
9) Podh NK#, Das A#, Dey P, Paliwal S, **Mehta G***. Single-molecule tracking for studying protein dynamics and target-search mechanism in live cells of *S. cerevisiae*. ***STAR Protocols (Cell Press)*** **2022**; 3(4): 101900, IF: 1.3, Q1
10) **Mehta G***, Sanyal K, Suman A, Eerappa R, Ghosh SK*. Minichromosome Maintenance Proteins in Eukaryotic Chromosome Segregation. ***BioEssays*** **2022***;* 44(1): 0265-9247, IF: 2.7, Q1
11) Podh NK, Paliwal S, Dey P, Das A, Morjaria S, **Mehta GD***. In-vivo Single-Molecule Imaging in Yeast: Applications and Challenges. ***Journal of Molecular Biology*** **2021**; 433(22):167250, IF: 4.5, Q1

Before joining IIT Hyderabad:

12) **Mehta GD**, Ball DA, Eriksson PR, Chereji RV, Clark DJ, McNally JG, Karpova TS. Single-molecule analysis reveals linked cycles of RSC chromatin remodeling and Ace1 transcription factor binding in yeast. ***Molecular Cell*** **2018**; 72:875-887.
13) **Mehta GD**, Anbalagan GK, Singh AP, Gadre P, Ghosh SK. An interplay between Shugoshin and Spo13 for centromeric cohesin protection and sister kinetochore monoorientation during meiosis I in *Saccharomyces cerevisiae*. ***Current Genetics*** **2018**; 64:1141-1152.
14) Ball D#, **Mehta GD#**, Salomon-Kent R, Mazza D, Morisaki T, Mueller F, McNally JG, Karpova T. Single-molecule tracking of Ace1p in *Saccharomyces cerevisiae* defines a characteristic residence time for non-specific interactions of transcription factors with chromatin. ***Nucleic Acids Research*** **2016**; 44(21);e160.

15) Agarwal M, **Mehta GD**, Ghosh SK. Role of Ctf3 and COMA subcomplexes in meiosis: implication in maintaining Cse4 at the centromere and numeric spindle poles. ***Biochimica Biophysica Acta-Molecular Cell Research* 2015**; 1853(3);671-684.
16) **Mehta GD**, Agarwal M, Ghosh SK. Functional characterization of kinetochore protein, Ctf19 in meiosis I: an implication of differential impact of Ctf19 on the assembly of mitotic and meiotic kinetochores in *Saccharomyces cerevisiae*. ***Molecular Microbiology* 2014**; 91(6):1179-1199.
17) **Mehta GD**, Kumar R, Srivastava S, Ghosh SK. Cohesin: Functions beyond sister chromatid cohesion. ***FEBS Letters* 2013**; 587(15): 2299-2312.
18) Das A, Kapoor A, **Mehta GD**, Ghosh SK, Sen S. Extracellular Matrix Density Regulates Extracellular Proteolysis via Modulation of Cellular Contractility. ***Journal of Carcinogenesis and Mutagenesis* 2013**; S13:003. doi: 10.4172/2157-2518.S13-003.
19) Lahiri S, **Mehta GD,** and Ghosh SK. Iml3p, a component of the Ctf19 complex of the budding yeast kinetochore, is required to maintain kinetochore integrity under conditions of spindle stress. ***FEMS Yeast Research* 2013**; 13(4): 375-385.
20) **Mehta GD**, Rizvi SM, and Ghosh SK. Cohesin: a guardian of genome integrity. ***Biochimica Biophysica Acta-Molecular Cell Research* 2012**; 1823(8): 1324-1342.
21) **Mehta GD**, Agarwal M, and Ghosh S. Centromere Identity: A challenge to be faced. ***Molecular Genetics and Genomics* 2010**; 284(2): 75-94.
22) Ray S, **Mehta GD** and Srivastava S. Label-free detection techniques for protein microarrays: prospects, merits and challenges. ***Proteomics* 2010**; 10: 731-748.
23) Singh AK, **Mehta GD** and Chhatpar HS. Optimization of medium constituents for improved chitinase production by *Paenibacillus sp. D1* using a statistical approach. ***Letters in Applied Microbiology* 2009**; 49: 708-714.

* Corresponding Author

Co-first author

AWARDS AND ACHIEVEMENTS (AFTER JOINING IIT HYDERABAD)

2025 Travel award to attend the FoundingGIDE Imaging Data Sharing meeting in Brisbane, Australia (https://founding-gide.eurobioimaging.eu/community-event-2025/), and to represent the initiatives of the India BioImaging community (www.indiabioimaging.org/).

2024 Faculty Teaching Excellence Award, IIT Hyderabad, India

2021 Ramalingaswami Fellowship, Department of Biotechnology, Govt. of India

2020 Har-Govind Khorana Innovative Young Biotechnologist Award, Department of Biotechnology, Govt. of India

SANCTIONED RESEARCH GRANTS (AFTER JOINING IIT HYDERABAD)

Sr. no.	Project title	Funding Agency	PI/ Co-PI	Project Duration From	To	Amount (INR in lakhs)
1	Single-molecule imaging of yeast sirtuin Sir2 to understand its trafficking and target-search	DBT	PI	May 2025	May 2028	78.57

	mechanism for transcription silencing.					
2	National Facility for Single-molecule and Super-Resolution Imaging	DBT-SAHAJ	PI	May 2024	May 2029	580.0
3	Role of chromatin remodelers in meiotic recombination and transcriptional switch during yeast meiosis, with emphasis on genetic disorders, infertility, and cancers.	Japan International Cooperation Agency (JICA)	PI	Jan 2023	Nov 2024	20.00
4	Mechanistic understanding of the functioning of CHD1 remodelers using biochemical, structural, and Single-Molecule Imaging approaches	DBT	Co-PI	Nov 2022	Nov 2025	54.76
5	Exploring the cohesin ring-independent functions of Rec8 during yeast meiosis.	DBT	PI	Aug 2021	Aug 2026	42.5
6	Elucidating the role of chromatin remodelers in yeast meiosis, especially in meiotic recombination, chromosome segregation and transcription of meiosis-specific genes.	DBT	PI	Aug 2021	Aug 2024	61.624
7	Single-Molecule Dynamics of the Cancer Therapy Target Aurora Kinase B	Seed Grant from IITH	PI	Apr 2021	Apr 2023	30.00

CONFERENCES (AFTER JOINING IIT HYDERABAD)

1) **Mehta GD.** Single-molecule tracking in live cells to quantify protein dynamics and the target search mechanism to understand the molecular mechanism of cell division. *Fluorescence Society XVI meeting*, Saha Institute of Nuclear Physics, Kolkata, Dec 2025. **(Invited Speaker)**

2) **Mehta GD**. The dynamic pool of Rec8-cohesin is crucial for meiotic recombination and transcription regulation in the yeast *S. cerevisiae*. *India Yeast Meeting*, IISc Bangalore, Dec 2025 **(Invited Speaker)**

3) **Mehta GD**. Dynamic pool of Rec8-cohesin is crucial for meiotic recombination and transcription regulation in the yeast *S. cerevisiae*. *9th ASIA Chromatin Meeting*, IISER Pune, Oct 2025 **(Invited Speaker)**

4) **Mehta GD.** Single-molecule tracking in live yeast *S. cerevisiae* reveals the dynamics of aurora kinase B (Ipl1) recruitment to the kinetochores and spindles. *Yeast2025*, Sorbonne University, Paris, July 2025 **(abstract selected for oral presentation)**

5) **Mehta GD**. Single-molecule imaging to unravel the dynamic regulation of aurora kinase B (Ipl1) in yeast *S. cerevisiae*. *Chromosome Stability Meeting*, JNCASR Bangalore, Dec 2024 **(Invited Speaker)**

6) **Mehta GD**. Exploring the cohesin ring-independent functions of cohesin subunits during yeast meiosis. *Meeting on DNA-Chromatin Fanatics*, IISER Pune, June 2024. **(Invited speaker)**

7) **Mehta GD**. Single-molecule tracking of cancer therapy target aurora kinase B to reveal its dynamics and target-search mechanism during cell division. *International Conference on Cancer Biology*, IIT Madras, India, Sept 2023. **(Invited speaker)**

8) **Mehta GD.** Single-molecule tracking of cancer therapy target aurora kinase B to reveal its dynamics and target search mechanism during cell division. *Yeast India 2023 meeting*, IISER Mohali, India, March 2023. (**Invited speaker**)

9) **Mehta GD**. Single-molecule tracking to understand the cross-talk between mitotic kinases and phosphatases to regulate cell division. *Chromosome Stability Meeting*, IISER Thiruvananthapuram, India. Dec 2022. (**Invited speaker**)

OUTREACH ACTIVITIES (AFTER JOINING IIT HYDERABAD)

2026 Organizing an **International Conference cum Workshop** on Single-Molecule Biophysics at IIT Hyderabad in January 2026. Conference website: www.simbio2026.com. >50% invited speakers are from abroad. The event is funded by the DST-JSPS bilateral seminar/symposia scheme, along with some industry sponsors and partners.

2025 Invited as a **resource person** for the Ashoka University-Zeiss Professional Course in Microscopy, Ashoka University, Haryana, India

2025 Invited as a **resource person** for the Microscopy and Image Analysis Training Course, IISER Pune, India.

2025 Organized an **online workshop** on 'Image Processing and Data Analysis'. Number of participants: 44 (including participants from Italy, Germany, and Malaysia), Revenue Generated from the registration fees: Rs. 1,92,900.

2024 Developed a 12-week **NPTEL course** on 'Advanced Fluorescence Microscopy and Image Processing'. This course was attended by 1575 students in 2025.

2024 Invited speaker for the **Faculty Development Program** (FDP) at the Amity University, Gurugram, India

2024 Invited speaker for the **Faculty Development Program** (FDP) at the National Institute of Technology, Warangal, India.

2024 Organized a skill development workshop at the Department of Biotechnology, IIT Hyderabad, on "**Artificial intelligence (AI) assisted manuscript writing**" (Guest speaker: Dr. Sowmiya Rani), for the students of IIT Hyderabad. Number of participants: 60, Registration fees: free

2024 Participated in the **Navodaya Teachers Training Program** at IIT Hyderabad, gave a talk and hands-on demonstration on visualizing cellular processes under the fluorescence microscope.

2024 Organized a workshop to enhance placements of the students of the Biotechnology Department, IIT Hyderabad, on "How to land your first biotech job?" (Guest Speaker: Dr. Vihang Ghalsasi), Number of participants: 60, Registration fees: free

2023-2024 Participated in the **Open to All Teaching (OAT)** initiative of IIT Hyderabad to teach the course BT6053 (Advanced Microscopy and Image Processing).

2023 Organized an online workshop **"Mei-India 2023"** to bring together all the scientists working in the field of meiosis in India. Number of participants: 40, Registration fees: free

2022 Invited speaker for the ‘High-End workshop on ultrastructural imaging and its applications in livestock research' at the NIAB, Hyderabad, under the **Karyashala Accelerate Vigyan scheme**.

2022 Invited to judge posters and panel discussion for the **HySci2022** at the Centre for DNA Fingerprinting and Diagnostics (CDFD), Hyderabad.

2022 Invited speaker for the **India Investigator Network (IIN)**.

STUDENT’S ACHIEVEMENTS

1. Mr. Nitesh Kumar Podh (bo21resch11008): Received Prime Minister’s Research Fellowship (PMRF) in 2021
2. Mr. Partha Dey (bo21resch11003): Received Prime Minister’s Research Fellowship (PMRF) in 2021
3. Ms. Rashmi Singh Yadav (bt24resch11009): Received DST-WISE Ph.D. scholarship
4. Mr. Nitesh Kumar Podh (bo21resch11008): Received best poster presentation award at the Chromosome Stability meeting at JNCASR, Bangalore, Dec 2024.
5. Mr. Partha Dey (bo21resch11003) was selected to attend the Bangalore Microscopy Course at NCBS Bangalore (Sept 2025) from an international competition.
6. Ms. Akriti Kumari’s (bo20resch11009) project was selected for funding under the BUILD (Bold and Unique Idea Led Development) program of IIT Hyderabad.
7. M.Tech student, Mr. Ayan Das (bo20mtech14004), was selected for the fully funded Ph.D. program at the University of New South Wales (UNSW), Sydney, Australia.
8. M.Tech student, Ms. Swarangi Tambat (BO22MTECH14002), was selected for the fully funded Ph.D. program at the University of Osaka, Japan, under the JICA FRIENDSHIP2 fellowship.
9. M.Tech student, Mr. Aditya Subramanian (bo21mtech14008), received a fully funded Ph.D. position at the University of Alabama, USA.
10. M.Tech student, Ms. Sarah Nawaz (bo21mtech11001), joined Thermo Fisher Scientific, Bangalore, as a Scientist I.
11. Ph.D. student, Mr. Nitesh Kumar Podh (bo21resch11008), received an opportunity for post-doctoral research at the University of Warwick, UK.